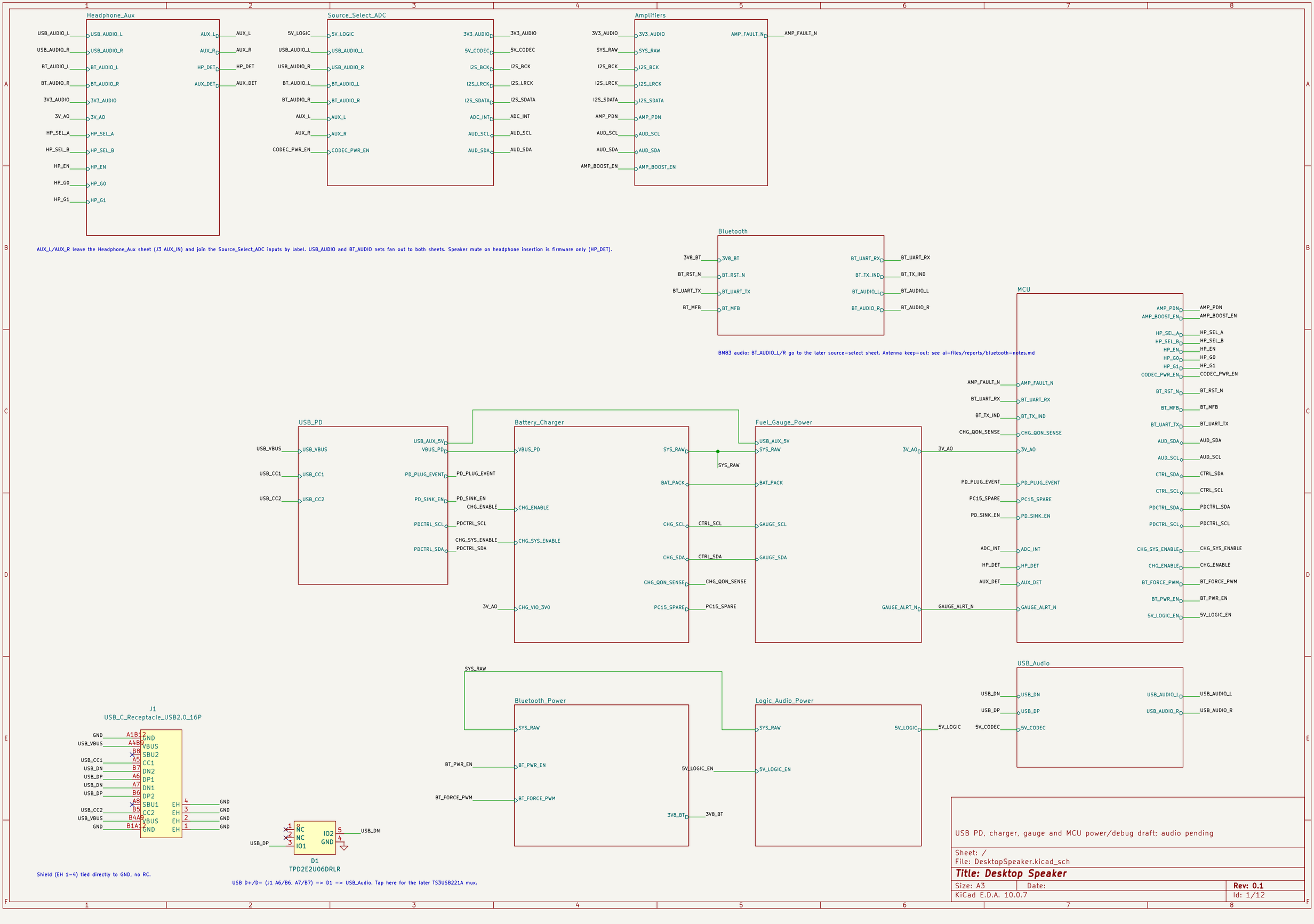
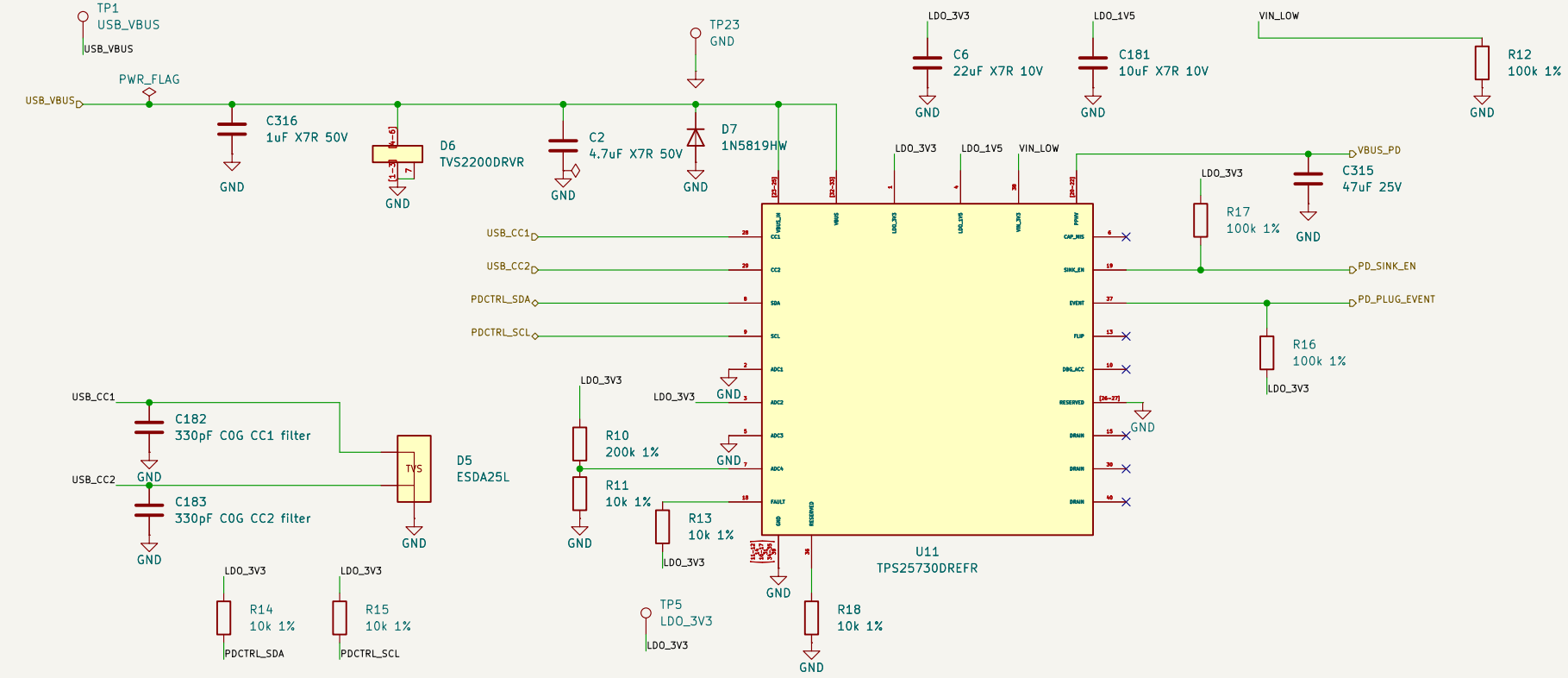

Headphone_Aux
USB_AUDIO_L
USB_AUDIO_R
BT_AUDIO_L
BT_AUDIO_R
3V3_AUDIO
3V_AO
HP_SEL_A
HP_SEL_B
HP_EN
HP_G0
HP_G1
AUX_L
AUX_R
HP_DET
AUX_DET
Source_Select_ADC
5V_LOGIC
CODEC_PWR_EN
5V_CODEC
I2S_BCK
I2S_LRCK
I2S_SDATA
ADC_INT
AUD_SCL
AUD_SDA
Amplifiers
SYS_RAW
AMP_PDN
AMP_BOOST_EN
AMP_FAULT_N
AUX_L and AUX_R leave the Headphone_Aux sheet (J3 AUX_IN) and join the Source_Select_ADC inputs by label. USB_AUDIO and BT_AUDIO nets fan out to both sheets. Speaker mute on headphone insertion is firmware only (HP_DET).
Bluetooth
3V8_BT
BT_RST_N
BT_UART_TX
BT_MFB
BT_UART_RX
BT_TX_IND
BT_AUDIO_L
BT_AUDIO_R
BM83 audio: BT_AUDIO_L/R go to the later source-select sheet. Antenna keep-out: see ai-files/reports/bluetooth-notes.md
MCU
CHG_QON_SENSE
PD_PLUG_EVENT
PC15_SPARE
PD_SINK_EN
GAUGE_ALRT_N
HP_SEL_A
HP_SEL_B
HP_EN
HP_G0
HP_G1
CODEC_PWR_EN
BT_RST_N
BT_MFB
BT_UART_TX
AUD_SDA
AUD_SCL
CTRL_SDA
CTRL_SCL
PDCTRL_SDA
PDCTRL_SCL
CHG_SYS_ENABLE
CHG_ENABLE
BT_FORCE_PWM
BT_PWR_EN
5V_LOGIC_EN
USB_PD
USB_VBUS
USB_CC1
USB_CC2
USB_AUX_5V
VBUS_PD
PD_PLUG_EVENT
PD_SINK_EN
PDCTRL_SCL
PDCTRL_SDA
Battery_Charger
CHG_ENABLE
CHG_SYS_ENABLE
CHG_VIO_3V0
SYS_RAW
BAT_PACK
CHG_SCL
CHG_SDA
CHG_QON_SENSE
PC15_SPARE
Fuel_Gauge_Power
GAUGE_SCL
GAUGE_SDA
3V_AO
J1
USB_C_Receptacle_USB2.0_16P
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
CC2
EH
D1
TPD2E2U06DRLR
NC
IO2
IO1
USB_DP
USB_DN
Shield (EH 1-4) tied directly to GND, no RC.
USB D+/D- (J1 A6/B6, A7/B7) -> D1 -> USB_Audio. Tap here for the later TS3USB221A mux.
Bluetooth_Power
Logic_Audio_Power
5V_LOGIC
USB_Audio
USB_DN
USB_DP
5V_CODEC
USB_AUDIO_L
USB_AUDIO_R
USB PD, charger, gauge and MCU power/debug draft; audio pending
Sheet: /
File: DesktopSpeaker.kicad_sch
Title: Desktop Speaker
Size: A3
Date:
Rev: 0.1
KiCad E.D.A. 10.0.7
Id: 1/12



5–20V SPR: ADCIN2=7 selects maximum per TI EVM Guide.

Read accepted PDO/RDO before raising charger/load limits; 5V default is not 2A permission.

Enforce 100mA SDP attach policy; implement USB suspend separately.

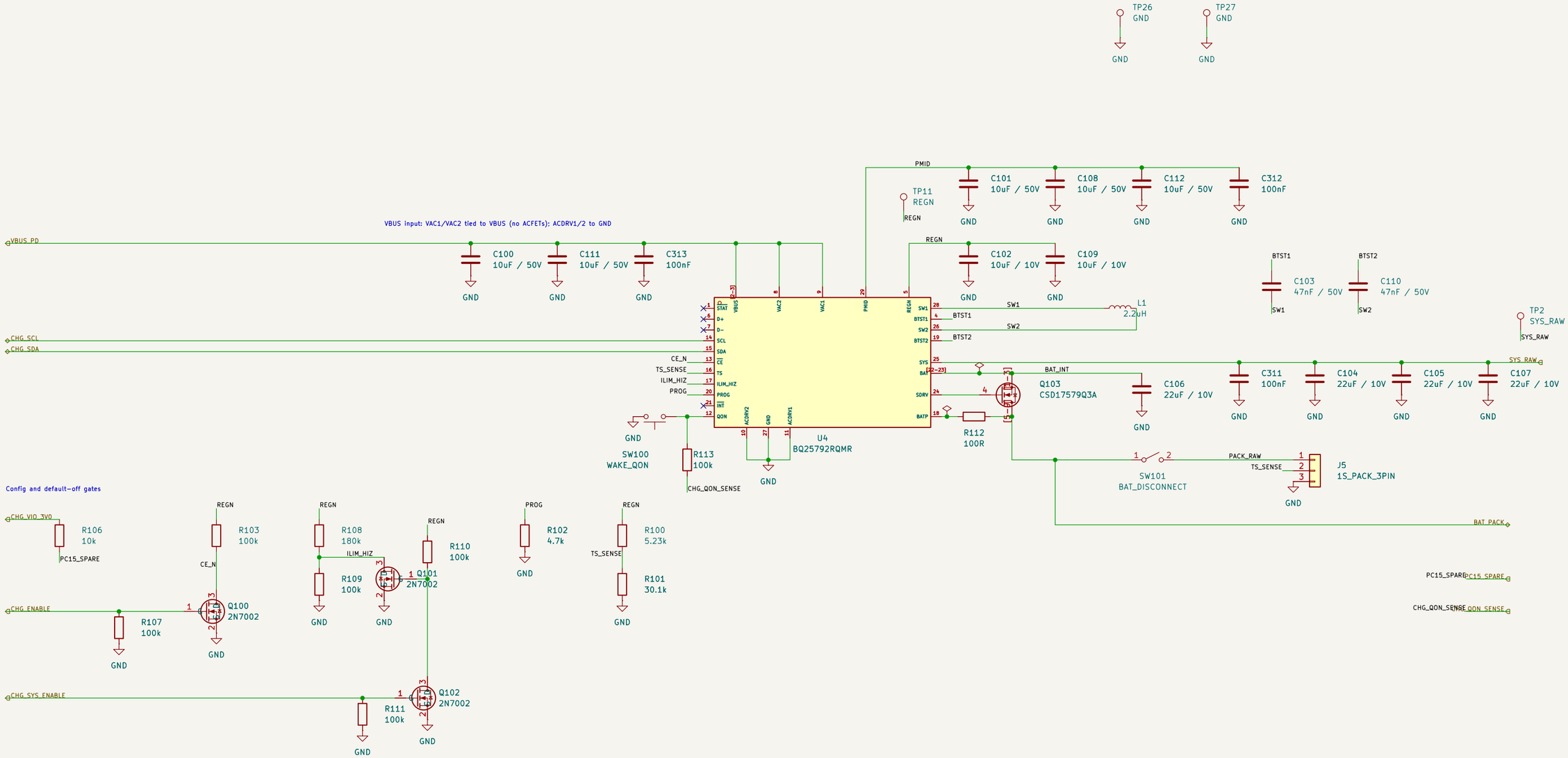
TVS2200 clamp max exceeds 28V abs max; qualify controller–pin transients.

Validate shared ~50uF bank bias and ramps; details: ai–files/reports/5–20v–pd–front–end–review.md.

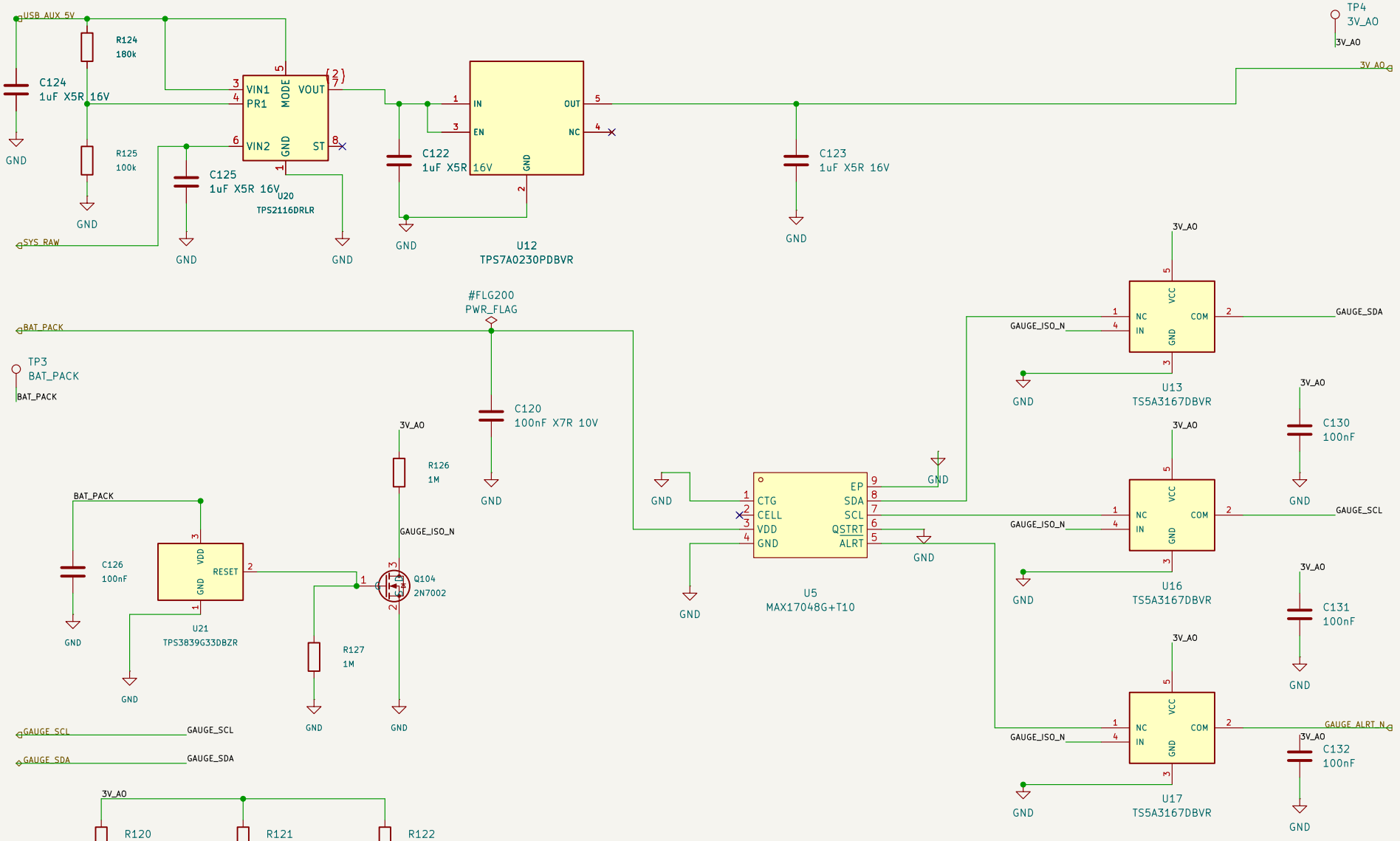
Sheet: /USB_PD/
File: USB_PD.kicad_sch

Title: USB–C PD 5-20 V SPR – TPS25730D candidate

Size: A4	Date:	Rev: prototype review
KiCad E.D.A. 10.0.7		Id: 2/12



Sheet: /Battery_Charger/	
File: Battery_Charger.kicad_sch	
Title: BQ25792 1S charger and power path candidate	
Size: A3	Date:
KiCad E.D.A. 10.0.7	Id: 3/12
	Rev: candidate-0.1



USB-priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus: AO-powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /Fuel_Gauge_Power/
File: Fuel_Gauge_Power.kicad_sch

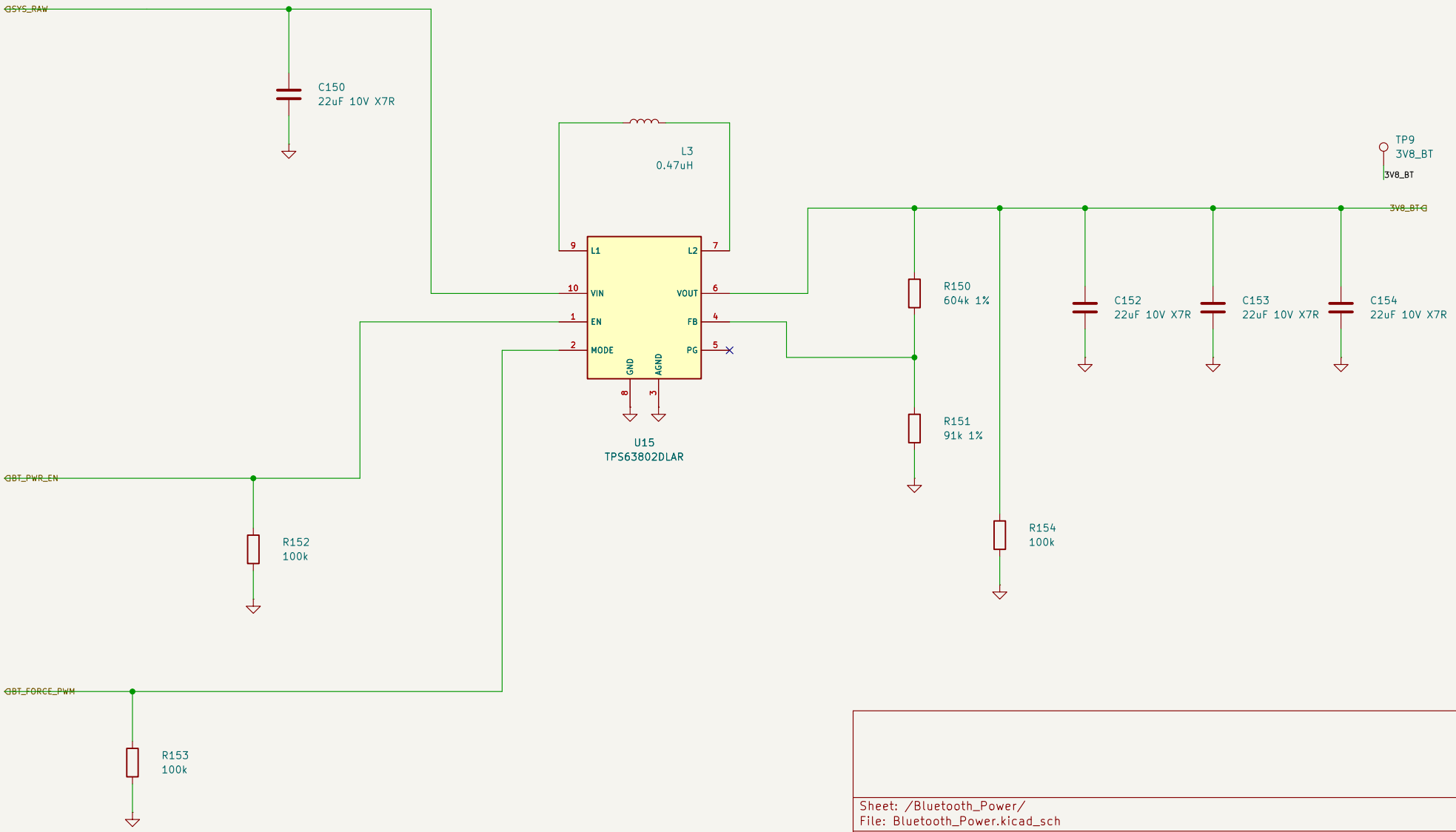
Title: Fuel gauge and low-current power

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.7		Id: 4/12

SYS to switched 3.8 V for BM83; reserve 500 mA. Module wiring follows later.

EN low disconnects output; default PFM. Local verified package footprints/STEP assigned.

Switch OFF after BM83 power-off ACK; validate >640 us supply decay and signal isolation.

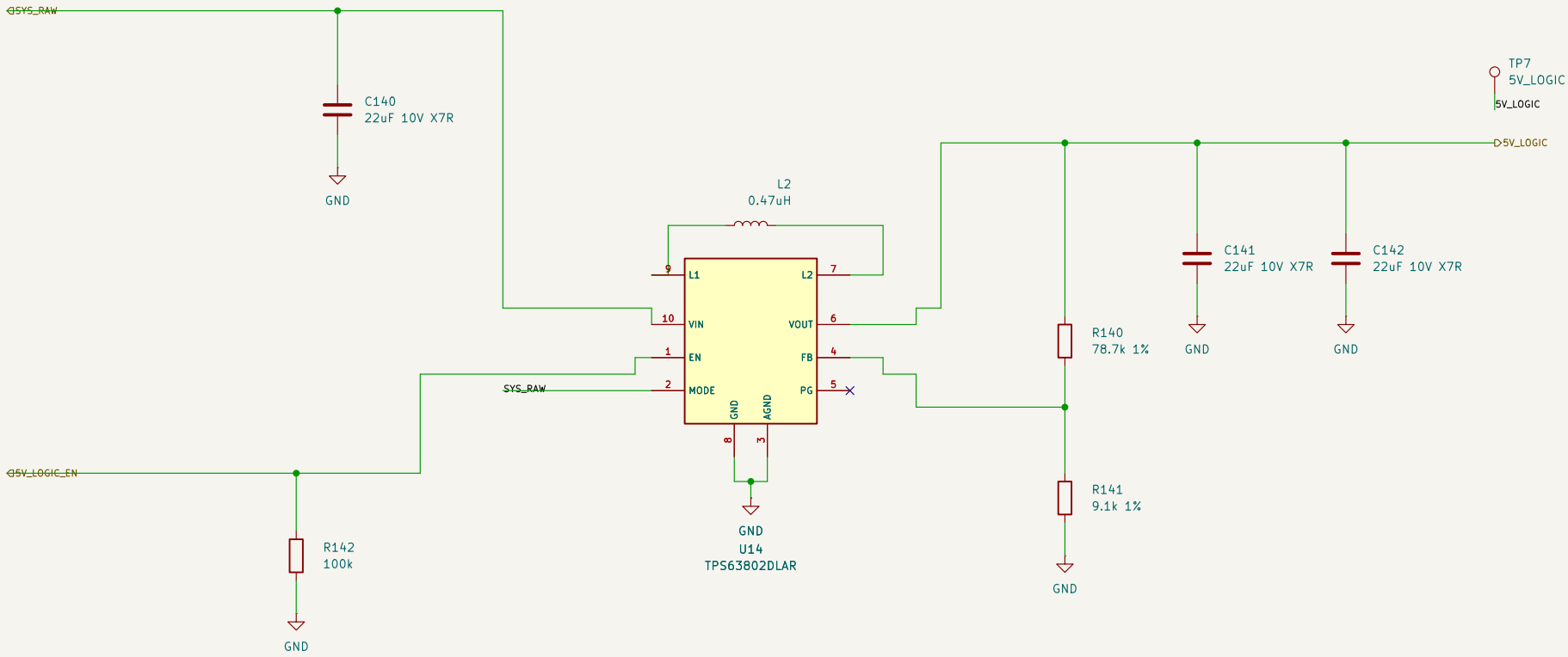


Sheet: /Bluetooth_Power/		
File: Bluetooth_Power.kicad_sch		
Title: Switched 3.8 V Bluetooth supply		
Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 5/12

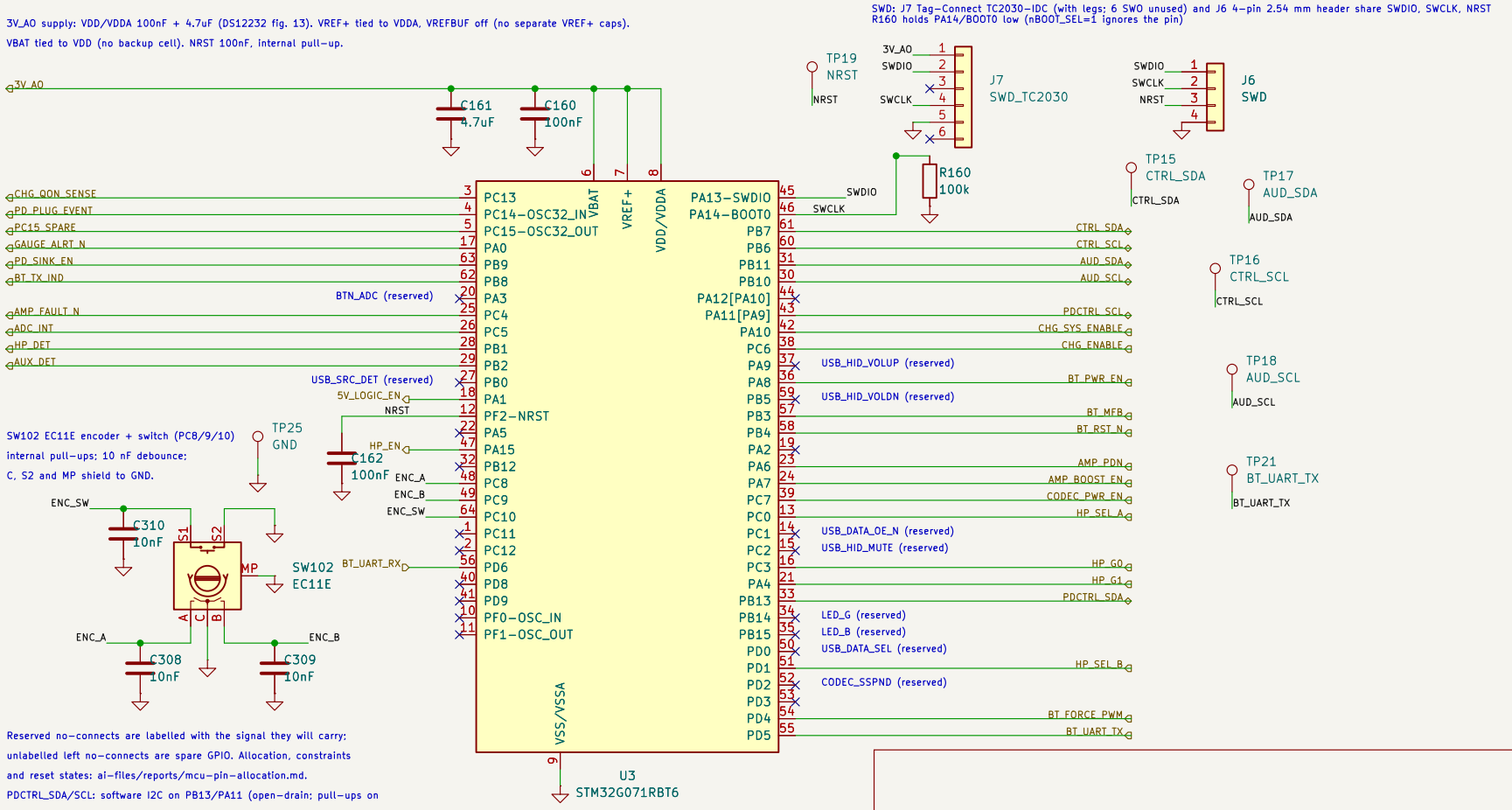
Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

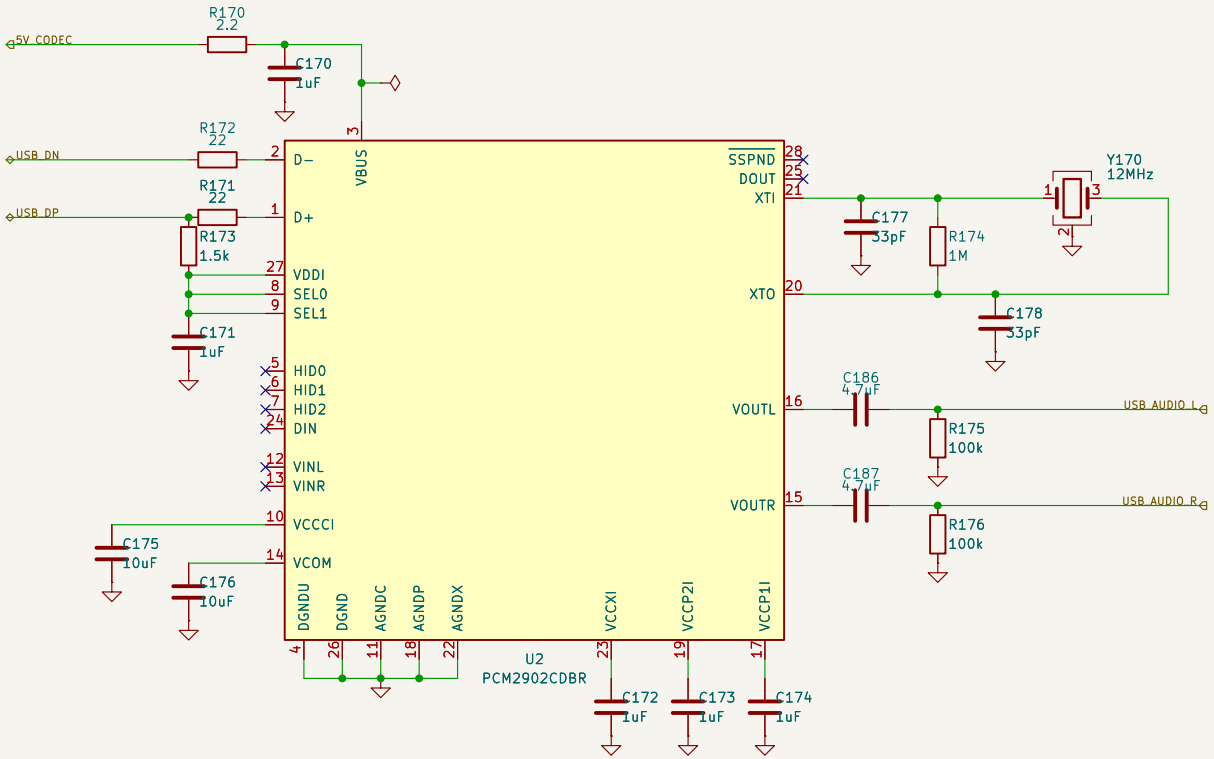
Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



Sheet: /Logic_Audio_Power/		
File: Logic_Audio_Power.kicad_sch		
Title: Regulated 5 V logic and audio supply		
Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 6/12


3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ tied to VDDA, VREFBUF off (no separate VREF+ caps).
VBAT tied to VDD (no backup cell). NRST 100nF, internal pull-up.
SWD: J7 Tag-Connect TC2030-IDC (with legs; 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)
3V_AO
C161 4.7uF
C160 100nF
TP19 NRST
J7 SWD_TC2030
J6 SWD
R160 100k
TP15 CTRL_SDA
TP17 AUD_SDA
TP16 CTRL_SCL
TP18 AUD_SCL
TP21 BT_UART_TX
CHG_QON_SENSE
PD_PLUG_EVENT
PC15_SPARE
GAUGE_ALRT_N
PD_SINK_EN
BT_TX_IND
BTN_ADC (reserved)
AMP_FAULT_N
ADC_INT
HP_DET
AUX_DET
USB_SRC_DET (reserved)
5V_LOGIC_EN
NRST
HP_EN
C162 100nF
ENC_A
ENC_B
ENC_SW
SW102 EC11E encoder + switch (PC8/9/10)
internal pull-ups: 10 nF debounce;
C, S2 and MP shield to GND.
TP25 GND
C310 10nF
C308 10nF
C309 10nF
SW102 EC11E
BT_UART_RX
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PA0
PB9
PB8
PA3
PC4
PC5
PB1
PB2
PB0
PA1
PF2-NRST
PA5
PA15
PB12
PC8
PC9
PC10
PC11
PC12
PD6
PD8
PD9
PF0-OSC_IN
PF1-OSC_OUT
VBAT
VREF+
VDD/VDDA
VSS/VSSA
PA13-SWDIO
PA14-BOOT0
PB7
PB6
PB11
PB10
PA12[PA10]
PA11[PA9]
PA10
PC6
PA9
PA8
PB5
PB3
PB4
PA2
PA6
PA7
PC7
PC0
PC1
PC2
PC3
PA4
PB13
PB14
PB15
PD0
PD1
PD2
PD3
PD4
PD5
SWDIO
SWCLK
CTRL_SDA
CTRL_SCL
AUD_SDA
AUD_SCL
PDCTRL_SCL
CHG_SYS_ENABLE
CHG_ENABLE
USB_HID_VOLUP (reserved)
BT_PWR_EN
USB_HID_VOLDN (reserved)
BT_MFB
BT_RST_N
AMP_PDN
AMP_BOOST_EN
CODEC_PWR_EN
HP_SEL_A
USB_DATA_OE_N (reserved)
USB_HID_MUTE (reserved)
HP_G0
HP_G1
PDCTRL_SDA
LED_G (reserved)
LED_B (reserved)
USB_DATA_SEL (reserved)
HP_SEL_B
CODEC_SSPND (reserved)
BT_FORCE_PWM
BT_UART_TX
U3
STM32G071RBT6
Reserved no-connects are labelled with the signal they will carry;
unlabelled left no-connects are spare GPIO. Allocation, constraints
and reset states: ai-files/reports/mcu-pin-allocation.md.
PDCTRL_SDA/SCL: software I2C on PB13/PA11 (open-drain; pull-ups on
the PD sheet). AUD_SDA/SCL: hardware I2C2, PB11/PB10 AF6.
CHG_QON_SENSE: PC13 (WKUP2), input only, high-impedance QON tap.
BT_RST_N: drive open-drain (low only). BT_MFB/BT_UART_TX: drive low
before BT_PWR_EN goes low (3V8_BT back-power).
BT_TX_IND = module P0_0 (EXTI/wake).
STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links
Sheet: /MCU/
File: MCU.kicad_sch
Title: MCU and programming
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.7
Id: 7/12

Supply: 5V_CODEC (U23 load switch from 5V_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
Codec is powered only when CODEC_PWR_EN (U23) and 5V_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC_PWR_EN on DCP/unknown sources.
SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



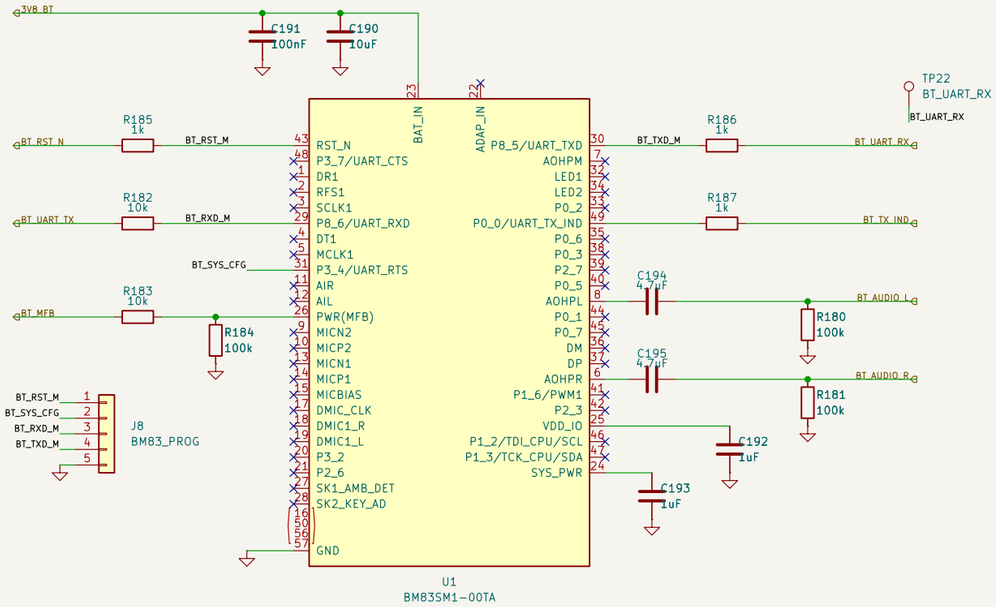
Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF ~0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D− path; analogue out to USB_AUDIO_L/R

Sheet: /USB_Audio/
File: USB_Audio.kicad_sch

Title: USB audio

Size: A4 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.7 | Id: 8/12



BM83SM1-00TA (DS70005402D). BAT_IN 3.2–4.2 V from 3V8_BT (U15); SYS_PWR/VDD_IO are outputs (decoupling only, do not load or feed). ADAP_IN unused (internal charger/USB DFU not used).

Host mode UART from the MCU: P8_6 RXD / P8_5 TXD, P0_0 = UART_TX_IND (wakes the MCU). MFB (PWR) = power-on/wake key, RST_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0_0 as TX_IND.

Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT_PWR_EN only after the module power-off ACK. Module outputs are 3.0–3.6 V into 5 V tolerant MCU pins.

Audio: single-ended DAC out (AOHPL/AOHPR) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.

Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.

J8 programming/test (isUpdate/Config Tool): 1 RST_N, 2 P3_4 SYS_CFG (low at reset = test mode), 3 UART_RXD (adapter TX), 4 UART_TXD (adapter RX), 5 GND. 3V8_BT must be enabled first.

BM83 module: host UART, MFB/RST_N, analogue out

Sheet: /Bluetooth/
File: Bluetooth.kicad_sch

Title: Bluetooth audio

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 9/12

Power: U22 TPS7A2033 (3V3_AUDIO, EN tied to IN) and U23 TPS22917 (5V_CODEC for USB_Audio, CODEC_PWR_EN, QOD tied to VOUT, CT open).
PCM1862: I2C mode (MD0 low), address 0x4A (MS/AD low), I2S master from the 24.576 MHz crystal (512 fs, PLL off), SCKI grounded. VIN1 = USB, VIN2 = BT, VIN3 = AUX.
Inputs: 2.2 uF – 100 R – 10 nF C0G (datasheet fig. 61 filter; -5 mA ESD-diode limit). AVDD via FB200. No reset pin: firmware re-initialises after every 3V3_AUDIO enable.
Crystal CL 15 pF (Luckl L3275240P11L): 2 x 22 pF with about 3 pF stray gives about 14 pF; trim at bring-up. Spare VIN4, MICBIAS and GPIO0/2/3 are no-connect.

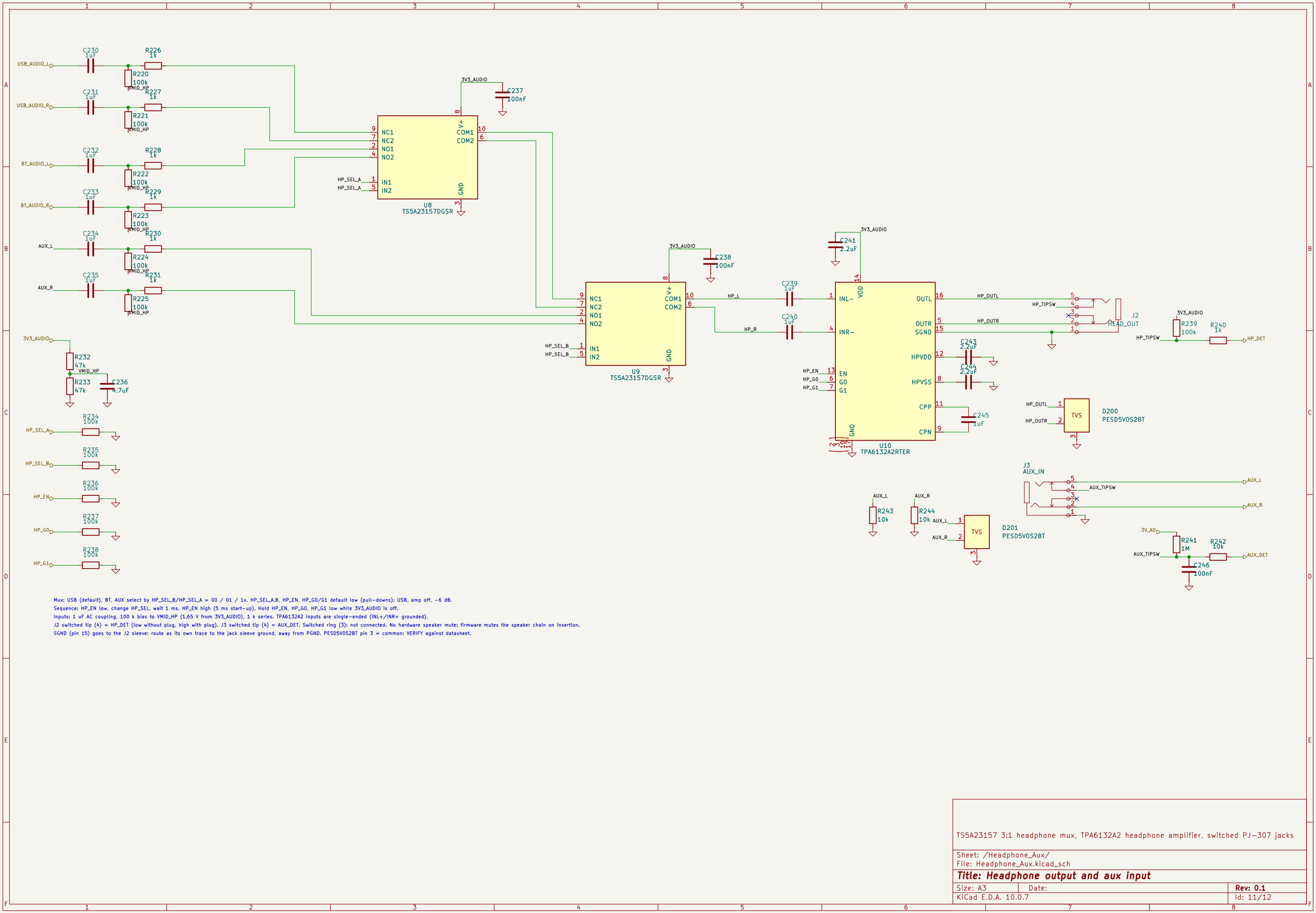
CODEC_PWR_EN
R212 100k
5V_LOGIC
U23 TPS22917DBVR
ON
CT
QOD
VIN
VOUT
GND
C224 1uF
C225 1uF
TP8 5V_CODEC
5V_CODEC
TP6 3V3_AUDIO
3V3_AUDIO
U22 TPS7A2033PDBVR
IN
EN
OUT
NC
GND
C221 1uF
C222 1uF
C223 10uF
FB200 BLM15AG601SN1D
C216 10uF
C215 100nF
C217 100nF
C218 10uF
C219 100nF
C220 1uF
U24 PCM1862DBTR
AVDD
DVDD
IOVDD
VINL1
VINR1
VINL2
VINR2
VINL3
VINR3
VINL4
VINR4
MICBIAS
MD0
MS/AD
SCKI
VREF
AGND
DGND
LDO
DOUT
BCK
LRCK
MC/SCL
MOSI/SDA
GPIO1/INTA
MISO/GPIO0
GPIO2/INTB
GPIO3/INTC
XI
XO
USB_AUDIO_L
USB_AUDIO_R
BT_AUDIO_L
BT_AUDIO_R
AUX_L
AUX_R
C200 2.2uF
C201 2.2uF
C202 2.2uF
C203 2.2uF
C204 2.2uF
C205 2.2uF
R200 100
R201 100
R202 100
R203 100
R204 100
R205 100
C206 10nF C0G
C207 10nF C0G
C208 10nF C0G
C209 10nF C0G
C210 10nF C0G
C211 10nF C0G
C212 1uF
C213 100nF
C214 10uF
C226 15pF C0G
C227 15pF C0G
24.576MHz Y200
R206 33
R207 33
R208 33
TP14 I2S_LRCK
I2S_LRCK
TP12 I2S_BCK
I2S_BCK
TP13 I2S_SDATA
I2S_SDATA
I2S_LRCK
I2S_BCK
I2S_SDATA
3V3_AUDIO
R209 2.2k
R210 2.2k
AUD_SCL
AUD_SDA
R211 100k
ADC_INT

PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch

Sheet: /Source_Select_ADC/
File: Source_Select_ADC.kicad_sch

Title: Source select and ADC

Size: A3 | Date: | **Rev: 0.1**
KiCad E.D.A. 10.0.7 | Id: 10/12



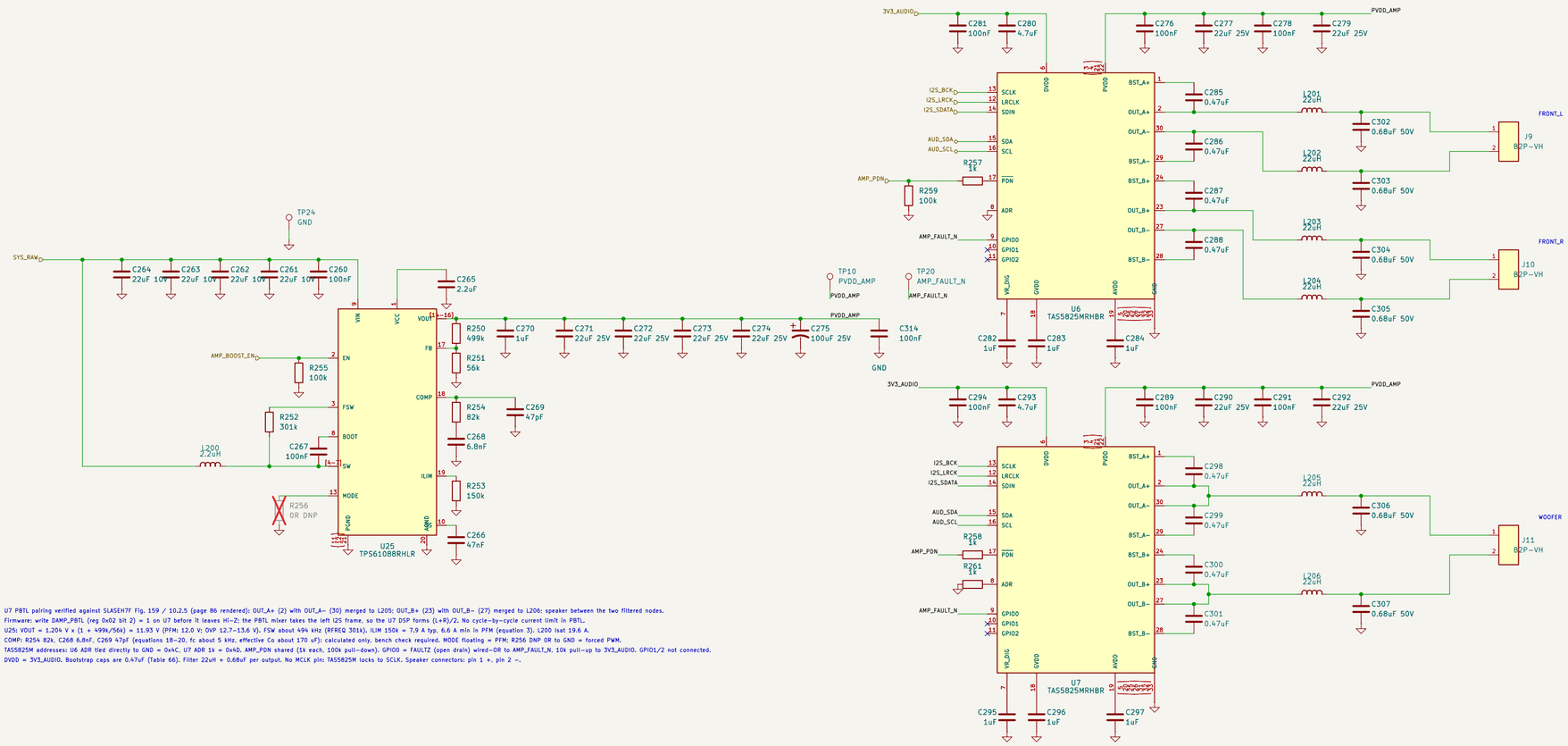
Mux: USB (default), BT, AUX select by HP_SEL_B/HP_SEL_A = 00 / 01 / 1x. HP_SEL_A,B, HP_EN, HP_G0/G1 default low (pull-downs): USB, amp off, −6 dB.
Sequence: HP_EN low, change HP_SEL, wait 1 ms, HP_EN high (5 ms start-up). Hold HP_EN, HP_G0, HP_G1 low while 3V3_AUDIO is off.
Inputs: 1 uF AC coupling, 100 k bias to VMID_HP (1.65 V from 3V3_AUDIO), 1 k series. TPA6132A2 inputs are single-ended (INL+/INR+ grounded).
J2 switched tip (4) = HP_DET (low without plug, high with plug). J3 switched tip (4) = AUX_DET. Switched ring (3): not connected. No hardware speaker mute: firmware mutes the speaker chain on insertion.
SGND (pin 15) goes to the J2 sleeve: route as its own trace to the jack sleeve ground, away from PGND. PESD5V0S2BT pin 3 = common: VERIFY against datasheet.

TS5A23157 3:1 headphone mux, TPA6132A2 headphone amplifier, switched PJ-307 jacks

Sheet: /Headphone_Aux/
File: Headphone_Aux.kicad_sch

Title: Headphone output and aux input

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.7		Id: 11/12



U7 PBTL pairing verified against SLASEH7F Fig. 159 / 10.2.5 (page 86 rendered): OUT_A+ (2) with OUT_A− (30) merged to L205; OUT_B+ (23) with OUT_B− (27) merged to L206; speaker between the two filtered nodes.
Firmware: write DAMP_PBTL (reg 0x02 bit 2) = 1 on U7 before it leaves HI-Z; the PBTL mixer takes the left I2S frame, so the U7 DSP forms (L+R)/2. No cycle-by-cycle current limit in PBTL.
U25: VOUT = 1.204 V × (1 + 499k/56k) = 11.93 V (PFM; 12.0 V: OVP 12.7–13.6 V). FSW about 494 kHz (RFREQ 301k), ILIM 150k = 7.9 A typ, 6.6 A min in PFM (equation 3). L200 Isat 19.6 A.
COMP: R254 82k, C268 6.8nF, C269 47pF (equations 18–20, fc about 5 kHz, effective Co about 170 uF) calculated only, bench check required. MODE floating = PFM; R256 DNP OR to GND = forced PWM.
TAS5825M addresses: U6 ADR tied directly to GND = 0x4C. U7 ADR 1k = 0x4D. AMP_PDN shared (1k each, 100k pull-down). GPIO0 = FAULTZ (open drain) wired-OR to AMP_FAULT_N, 10k pull-up to 3V3_AUDIO. GPIO1/2 not connected.
DVDD = 3V3_AUDIO. Bootstrap caps are 0.47uF (Table 66). Filter 22uH + 0.68uF per output. No MCLK pin: TAS5825M locks to SCLK. Speaker connectors: pin 1 +, pin 2 −.

TPS61088 PVDD boost, two TAS5825M (stereo BTL and mono PBTL), LC output filters, speaker connect

Sheet: /Amplifiers/		
File: Amplifiers.kicad_sch		
Title: Amplifiers		
Size: A2	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 12/12